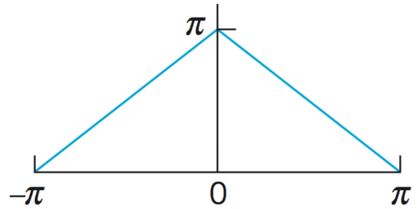
Please hand in the following problems:

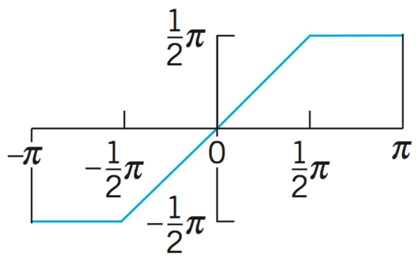
Problem 1: Determine the Fourier Series for each of the periodic functions depicted in the graphs below. Also, assume they periodically extend along the x-axis.

(i)



(ii)

(iii)



Problem 2: The point of this problem is to show that the two forms of the Fourier Series and Coefficients presented in class are equivalent. That is,

$$f(x) = \sum_{n=-\infty}^{\infty} d_n\, e^{inx} \qquad \text{where} \qquad d_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, e^{-inx}\, dx$$

is the same as computing

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx)\right)$$

$$\text{where} \quad a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx \qquad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\, dx \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\, dx$$

Proof:

$$\begin{aligned}\sum_{n=-\infty}^{\infty} d_n\, e^{inx} &= d_0 + \sum_{n=-\infty}^{-1} d_n\, e^{inx} + \sum_{n=1}^{\infty} d_n\, e^{inx} \\ &= [Details] \\ &= \underbrace{d_0}_{a_0} + \sum_{n=1}^{\infty}\Big[\underbrace{(d_{-n} + d_n)}_{a_n}\, \cos nx + \underbrace{(-d_{-n} + d_n)i}_{b_n}\, \sin nx\Big]\end{aligned}$$

Answer the following:

(a) Briefly explain or show why the first equality is true.

(b) Show the $[Details]$ above.

(c) Show that (i) $d_0 = a_0$, (ii) $d_{-n} + d_n = a_n$, (iii) $(-d_{-n} + d_n)i = b_n$,